Problem 15 | Problems T5_MechanicalProperties.pdf #15 | Brinell hardness · inverse problem

Indentation Diameter that Gives a Brinell Hardness of 300

Original statement · Problems T5_MechanicalProperties.pdf, p. 5

15. What will be the diameter of the indentation that produces a Brinell hardness of 300 when a load of 500 kg is applied? Diameter of the ball: 10 mm.

Official Solution: $d = 1.453$ mm

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. Same definition as in Problem 14: $HB = P/S$, with S the area of the spherical cap, $HB = \dfrac{2P}{\pi D\left(D - \sqrt{D^2 - D_i^2}\right)}$ [Session 12 Slide 19].
2. P in kilogram-force, lengths in mm; the hardness number has no unit attached.
3. Only the physically meaningful root is retained: the indentation diameter is positive and not larger than the ball, $0 < D_i \leq D$.

Data

Quantity	Symbol	Value
Brinell hardness	HB	300
Load	P	500 kg
Ball diameter	D	10 mm

Unknown: the diameter of the indentation D_i (called d in the key).

PHASE 2 · FRAMES & STRATEGY

This is the *inverse* of Problem 14: the hardness is known, and the indentation is the unknown. The formula of the slide is solved for D_i before substituting numbers:

1. Isolate the bracket: $D - \sqrt{D^2 - D_i^2} = \dfrac{2P}{\pi D\, HB} \equiv k$.
2. Isolate the square root: $\sqrt{D^2 - D_i^2} = D - k$.
3. Square both sides, $D^2 - D_i^2 = (D-k)^2$, and solve: $D_i^2 = D^2 - (D-k)^2 = 2Dk - k^2 = k\,(2D - k)$.

The squaring step is allowed here because $D - k > 0$ (the left side is a square root, and k is a small fraction of D), so no spurious root is introduced.

PHASE 3 · STEP-BY-STEP DERIVATION

1. The bracket k

$$k = D - \sqrt{D^2 - D_i^2} = \frac{2P}{\pi D\, HB} = \frac{2 \times 500}{\pi \times 10 \times 300} = \frac{1000}{9424.78} = 0.106103 \text{ mm}$$

2. The indentation diameter

$$D_i^2 = k\,(2D - k) = 0.106103 \times (20 - 0.106103) = 0.106103 \times 19.893897 = 2.11081 \text{ mm}^2$$

$$D_i = \sqrt{2.11081 \text{ mm}^2} = \mathbf{1.4529 \text{ mm}}$$

3. Verification by substitution

$\sqrt{D^2 - D_i^2} = \sqrt{100 - 2.11081} = 9.893897$ mm, which is $D - k = 10 - 0.106103 = 9.893897$ mm, as imposed in step 2. Putting $D_i = 1.453$ mm back in the formula:

$$HB = \frac{2 \times 500}{\pi \times 10 \times (10 - 9.893897)} = \frac{1000}{\pi \times 10 \times 0.106103} = 300.00$$

which recovers the 300 of the statement.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** k is a length (kg over kgf/mm² gives mm² and, divided by mm, a length); $D_i^2 = k(2D - k)$ is in mm².
- **Load factor.** $Q = P/D^2 = 500/10^2 = 5$, a standard value of [Session 12 Slide 20] for the ball of 10 mm.
- **Limit.** Since $k \ll D$ here, $D_i^2 \approx 2Dk = 4P/(\pi\, HB)$, so $D_i \approx \sqrt{4 \times 500/(\pi \times 300)} = 1.4567$ mm: the approximation is within 0.3 % of the exact value (1.4529 mm).
- **Direction.** A harder material (larger HB) leaves a smaller indentation under the same load: $D_i \propto HB^{-1/2}$ for small indentations.

Interpretation

A hardness of 300 HB under 500 kg leaves an indentation of only about 1.45 mm in a ball of 10 mm, with a depth of $h = k/2 = 0.05305$ mm. The ratio $D_i/D = 0.145$ is a very small indentation, which is difficult to measure accurately with a microscope (extension: the usual range of validity of the Brinell test is $0.25D$ to $0.6D$, a rule that is not in the slides).

Check against the official solution

Item	Official	Computed	Comment
indentation diameter	1.453 mm	1.4529 mm	agrees (the key rounds to three decimals)